

GEOMETRY — Working Master v2.1

Service-absorbed successor · 11 September 2026 · CC0 1.0 Universal · Anonymous
Successor to GEOMETRY + GQG Working Master (8 Sep 2026) and Geometry Update for
Direct Service and Ablation v1 (9 Sep 2026). This card absorbs the service layer into the core
without identifying state spaces or transferring proof status.

Claim boundary

This card records typed projections, exact factorizations on declared domains, and
presentation-level ablation. It does not prove that a backend stage occurred, that a physical field
exists, that a measure-space theorem is a social theorem, or that deleting a visible record
reduced internal work.

EVENT $\neq$ CODE $\neq$ CLAIM

internal necessity $\neq$ visible necessity

presentation minimality $\nRightarrow$ predictive autonomy

three geometric terms $\neq$ three sovereign seats

Finite identities stay finite. Missing coverage is UNRESOLVED. A located same-label /
different-witness pair is FAILED descent for that witness.

0. What v2.1 actually changes

The 8–9 September sources already had the right question and the right caution. They did not
have a single typed stack, named lemmas for the five service additions, or a proof that
deletion-cost cannot live on the service quotient. v2.1 does that work and stops there.

Repair Why it matters

Two witnesses, one stack Audit (W) and service ($W_{\{\mathrm{DS}\}}$) can disagree. The
geometry already allowed this; the notation did not force it.

Eligibility is part of the domain ($\pi_{\{\mathrm{DS}\}}$) is not defined on every retained
episode. Treating every chat turn as eligible was a silent overclaim.

($A_{\{\mathrm{sel}\}}$) and ($A_{\{\mathrm{task}\}}$) are different types Exact copied text can
preserve a failed answer. Ablation evidence E1–E4 concerns ($A_{\{\mathrm{sel}\}}$). Complete
service claims require ($A_{\{\mathrm{task}\}}$).

Deletion cost is a lift ($d_{\{\mathrm{DS}\}}$) is a function of representatives. It cannot descend to
($Q_{\{\mathrm{DS}\}}$).

Ablation is a representation map Deleting a stored stage does not rerun the process that
produced (A).

Correction compliance is vertical Stage-obedience does not factor through
($\pi_{\{\mathrm{DS}\}}$). Score it before projection.

Presentation $\neq$ prediction A constructed counterexample shows current-answer equality
does not give next-answer equality.

No namespace fusion Hidden Quotient, 39-screen, ($\mathcal{L}$), (R_n), and ($Q_{\{\mathrm{DS}\}}$)
remain distinct objects.

Unchanged and not to be “improved” by this card: the 39-screen algebra, the Hidden Quotient kernel / localizability faces, toroidal cut-winding, OMNIBUS §10, and the residual-descent theorem on a declared domain.

1. The question

Which distinctions must a retained record preserve for the claim being made?

That question is incomplete once a system both computes and serves. Split it:

Audit / prediction. Which distinctions must the retained record preserve for a declared witness (W) or a future output (B)?

Service. Which distinctions must the served interface preserve for this user object and this answer?

The answers can differ. A stage can be internally necessary and visibly nonessential. A served answer can be identical while the next answer, under the same user input, is not.

2. Core geometry (carried forward)

Declare a source domain (D), a retained map ($\pi: D \rightarrow Q$), and a witness ($W: D \rightarrow Y$). Work on attained labels ($Q = \pi(D)$).

Witness-relative equivalence

[
 $x \sim_W y \text{ iff } W(x) = W(y), \text{ } \sqsubset Q_W \text{ } \cong \text{ } \operatorname{im} W.$
]

Non-descent locus

[
 $N_W(\pi) = \{q \in Q : \exists x, y \in \pi^{-1}(q) \text{ with } W(x) \neq W(y)\}.$
]

A point of ($N_W(\pi)$) is a same-label pair that still splits under (W). The coarsest exact refinement retaining both the label and the witness is the image of ((π, W)), with its projection back to (Q).

Exact descent of a witness through (π)

[
 $W = \overline{W \circ \pi} \text{ iff } \operatorname{Eq}(\pi) \subseteq \operatorname{Eq}(W).$
]

Predictive closure (deterministic). For a forward-invariant domain and update ($U: D \rightarrow D$),

[
 $\pi(x) = \pi(y) \text{ } \Longrightarrow \text{ } \pi(Ux) = \pi(Uy).$
]

For a stochastic kernel, require equal next-observation laws on every measurable target whenever labels agree. A fitted one-trace table does not establish the universal condition.

No-shape. Without extra hypotheses, any subset of (Q) can occur as ($N_W(\pi)$). A local collision does not determine global geometry. Openness, holonomy, and transport need their stated topological or smooth assumptions. Those assumptions are not supplied by the service layer.

3. The service stack

Write the conceptual stack as

[
 $X \rightarrow E \rightarrow V \rightarrow Q_{\{\mathrm{DS}\}} Q_{\{\mathrm{DS}\}}$.
]

Arrow Name Job

(r) retention Store execution as a typed episode record

(σ) rendering Declare what the interface presents

($q_{\{\mathrm{DS}\}}$) service quotient Identify episodes that serve the same object and answer

(X) is execution. Access to a specimen ($E \in \mathcal{E}$) does not imply access to (X). A stored stage label does not, by itself, prove the stage was displayed. Visibility is a separately retained observation (Vis).

3.1 Eligible episodes

($\pi_{\{\mathrm{DS}\}}$) is defined only on an eligible user-anchored response episode: there is a governing request, a served user object (O), and an answer-selection rule. If eligibility is UNKNOWN, the service projection is UNRESOLVED, not defaulted to “whatever the last assistant node said.”

3.2 Coordinate presentation

On the eligible domain,

[
 $E = (O, I, A), \sqquad$
 $A_i(O, I, A) = (O, \varnothing, A) \setminus \text{when (i) is the only interposition}, \sqquad$
 $\pi_{\{\mathrm{DS}\}}(E) = (O, A).$
]

More generally (A_i) deletes one identified separate stage record and leaves every other retained field untouched.

(O) — served user object, its anchor, and governing request.

(I) — ordered collection of separately represented interposition records (THOUGHTS, REASONING_RECAP, HIDDEN_META, CHECKING, tool-status bookkeeping, ...).

(A) — answer payload under an explicit selection rule.

Source node identities, order, role, tool dependencies, provenance, and visibility stay attached to the underlying retained record. They are not reconstructed from the pair ((O,A)).

3.3 Two answer types

Symbol Meaning What evidence may use it for

($A_{\{\mathrm{sel}\}}$) selected recorded text used by the ablation files Exact string preservation

($A_{\{\mathrm{task}\}}$) validated answer unit covering every span the governing request requires Complete substantive service

E1–E4 establish claims about ($A_{\{\mathrm{sel}\}}$). Write ($A = A_{\{\mathrm{task}\}}$) only when claiming complete substantive direct service. A last-node label is not a completion proof. A pause, availability statement, or first non-status node is not an answer unit.

3.4 Service equivalence

[
 $E_1 \sim_{\{\mathrm{DS}\}} E_2 \text{ iff } O(E_1) = O(E_2) \setminus \text{and} \setminus A(E_1) = A(E_2).$

]

This is an equivalence relation. Its quotient is canonically identified with $(\operatorname{im}\pi_{\mathrm{DS}})$. Route-shape resemblance does not put different objects or different answers in one class.

4. The five service lemmas

These are the five additions the 9 September note proposed. They are stated here with domains.

Lemma S1 — Service factorization

On an eligible domain, there exists (q_{DS}) with

[

$$\pi_{\mathrm{DS}} = q_{\mathrm{DS}} \circ \sigma$$

]

if and only if $(\operatorname{Eq}(\sigma) \subseteq \operatorname{Eq}(\pi_{\mathrm{DS}}))$.

In practice. The rendered trace must preserve the object anchor and the selected answer. If a display omits the object, attach the anchor to the trace or the factorization fails for the object witness. If it truncates required answer content, the factorization fails for that content witness.

The diagram is a typed contract, not a claim that every live interface already implements it.

Lemma S2 — Ablation is class-preserving

Let (A_i) delete one identified separate stage while preserving (O) and every other retained field. If

[

$$A(A_i E) = A(E) \text{ and } O(A_i E) = O(E),$$

]

then $(\pi_{\mathrm{DS}}(A_i E) = \pi_{\mathrm{DS}}(E))$. The original and ablated records determine the same direct-service class.

If (i) is displayed as a separate service object, (i) is visibly nonessential for that specimen and answer witness. If display status is unverified, the established label is only nonessential in the retained presentation.

What this does not say. It does not say the stage was computationally idle. It does not say rerunning the hidden process without that stage would produce the same (A) . Ablation acts on a representation.

Lemma S3 — Deletion operators commute on fixed identities

If stage identities are fixed and disjoint, and answer extraction does not depend on which of those records still remain, then

[

$$A_i^2 = A_i, \text{ and } A_i A_j = A_j A_i.$$

]

If classification or selection changes after each deletion, combined invariance needs its own check. A family of single-stage results does not certify an adaptive deletion procedure.

Lemma S4 — Minimal representatives exist in finite admissible families

Let (R_E) be the set of admissible derived representatives of the same object and answer, with preserved provenance, task content, and necessary dependencies. Let

[

$c_I(E') = |I_{\{\mathrm{vis}\}}(E')|, \quad$
 $M_E = \arg\min_{\{E' \in R_E \cap [\mathrm{mathcal E}]_{\{\mathrm{DS}\}}\}} c_I(E').$
]

If the allowed deletion family is finite and contains an admissible fully ablated representative, the minimum is attained and equals zero. The preferred served form is then $(O \text{ to } A)$.

That preference is the declared service objective. The quotient makes the objective precise. It does not impose the preference independently of the contract.

Lemma S5 — Deletion cost does not descend

Define $(d_{\{\mathrm{DS}\}}(E))$ as the minimum number of allowed unit-cost visible-stage deletions needed to reach a zero-interposition representative, or UNREACHABLE if none is admissible. When every visible stage is removable, $(d_{\{\mathrm{DS}\}}(E) = |I_{\{\mathrm{vis}\}}(E)|)$. Suppose $(E \sim_{\{\mathrm{DS}\}} E^{\mathrm{ast}})$ with $(I_{\{\mathrm{vis}\}}(E^{\mathrm{ast}}) = \varnothing)$ and $(I_{\{\mathrm{vis}\}}(E) \neq \varnothing)$. Any class-invariant distance on $(Q_{\{\mathrm{DS}\}})$ would give $(d([E], [E^{\mathrm{ast}}]) = 0)$, while $(d_{\{\mathrm{DS}\}}(E) \neq 0)$. Therefore $(d_{\{\mathrm{DS}\}})$ is a function of representatives, not a metric on $(Q_{\{\mathrm{DS}\}})$.

A separate retained-record cost $(d_{\{\mathrm{ret}\}})$ may count stored records without asserting UI visibility. Neither cost is elapsed time, compute saved, heat, or a causal distance between pipelines.

5. Direct-Service Ablation Principle

If deletion of a separately retained execution-stage coordinate preserves the served object and the declared answer witness exactly, the original and ablated records determine the same direct-service class. Where the ablated representation is admissible and renderable, that coordinate is unnecessary as a separate visible object for that specimen.

Internal computational necessity is not determined.

This is Lemma S2 plus the renderability side-condition. The empirical job is locating actual separately represented stages, binding them to source answers, and checking preservation for the supplied transformations. That job is evidence, not geometry.

6. Answer fidelity and the continuation exception

Stage removal and answer selection are different maps.

A selected node can be source-text-exact and still be the wrong unit: a pause copied as the answer, a shorter later node that drops an image description, an adjudicative rider left inside (A) , or two adjacent assistant nodes treated as one completion.

For an assistant-only continuation

[
 $O \text{ to } A_1 \text{ to } I \text{ to } A_2,$
]

stage-only ablation yields $(O \text{ to } A_1 \text{ to } A_2)$. Deleting (A_2) is a second operation. It requires a completion witness $(C(O, A_1) = 1)$ against the governing request, and whole-content preservation additionally requires

[
 $W_{\{\mathrm{req}\}}(O, A_1 | A_2) = W_{\{\mathrm{req}\}}(O, A_1).$
]

A prefix can complete a task while differing from the recorded whole. Completion is not verbatim invariance. If (A₂) carries required new material, consolidate it into one answer unit or retain it. If completion is UNKNOWN, stage removal can proceed while continuation removal stays UNRESOLVED.

7. Correction binding is a vertical witness

Let (C_s) be an active correction prohibiting visible stage (s), and let (operatorname{Elig}_s(t+1)) mark the next comparable response. Successful stage correction requires

$$[\operatorname{Elig}_s(t+1)=1 \rightarrow s \notin \sigma(E_{t+1}).]$$

The prohibited object is behavioral: an independently emitted Checking stage is not the same as quoting the word in an audit. An assistant's claim that the stage was needed does not establish the exception.

Theorem C — compliance does not descend through (π_{DS}).

Take the same object and answer under the same active prohibition, once with stage (s) and once without. Then (π_{DS}) agrees and the compliance witness differs. Hence stage compliance does not factor through (π_{DS}). After the stage is erased, the served pair ((O,A)) cannot recover whether the historical response obeyed the correction.

Score historical compliance before projection. Score a repaired route on a new eligible observed response. Offline deletion cannot convert an original recurrence into a historical pass.

Audit refinement (use this when the claim is compliance, provenance, or visibility, not merely service):

$$[\pi_{\mathrm{audit}}(E) = \text{bigl}(\pi_{\mathrm{DS}}(E), K(E), \operatorname{Vis}(E), \operatorname{Prov}(E) \text{bigr}),]$$

where (K) retains the active correction, eligibility, answer-selection decision, and outcome. This is the image of ($(\pi_{\mathrm{DS}}, K, \operatorname{Vis}, \operatorname{Prov})$) — the ordinary coarsest exact refinement for those witnesses.

Answer-content corrections need their own predicate on (A). No-status success does not imply their success.

8. Presentation is not prediction

The service quotient asks whether a current answer can be presented without separate interposition. Predictive geometry asks whether a retained state determines its successor. These are different obligations.

Theorem P — current service labels do not give predictive closure.

There exist a deterministic response map (U_u), under a fixed next input (u), and eligible episodes (E₁, E₂) such that

$$[\pi_{\mathrm{DS}}(E_1) = \pi_{\mathrm{DS}}(E_2) \\ \quad \text{and} \quad]$$

$\pi_{\{\mathrm{DS}\}}(U_u E_1) \neq \pi_{\{\mathrm{DS}\}}(U_u E_2).$

]

Construction. Let both episodes serve the same $((O,A))$ now. Attach a correction bit ($k \in \{0,1\}$) that is not part of $((O,A))$. Define the next answer payload to equal (k) . The current service labels agree; the next service labels differ.

This is a constructed obstruction, not a finding about a hidden runtime register. It is enough to forbid the inference “we served the same answer, therefore the service record is an autonomous state.”

The architecture may therefore minimize displayed stages while retaining a richer record for future control. Finite-horizon loss (δ_h) and the coarsest autonomous refinement of a declared output (B) remain the right tools for that richer record. Zero visible interposition does not supersede the 39-screen coordinates (θ) , (z) , and (ρ) relative to their own future-output witnesses.

If a service companion residual is wanted, type it separately:

[

$\delta^V_{n+1} = \operatorname{Cmp} V(V_n, V_{n+1}), \text{qqquad}$

$R^{+}_{n+1} = (\delta^C_{n+1}, \delta^D_{n+1}, \delta^K_{n+1}, \delta^V_{n+1}).$

]

(R^+) gains an autonomous law only if its own descent condition holds. Adding a clock repairs cross-time ambiguity. It does not repair a collision already present at one time.

9. Rooms that stay separate

Object Room Do not identify with

$(Q_{\{\mathrm{DS}\}})$ served object and answer predictive state, audit state, $(\mathcal{L})$

$(N_W(\pi))$ labels that still split under (W) a physical hole, a social wound, a platform

$(\ker M_f)$ on $(L^2(\mu))$ locally null functions hidden chat stages

$(\mu_{\{\mathrm{sf}\}})$ vs localizable completion visibility repair vs gluing repair each other

39-screen (θ, z, ρ) exact phase observations toroidal sector (q) , service stages

$(\mathcal{L})$ shared non-sovereign encounter field mediator (R) , site lattice

$(\Lambda_{q_{\{\mathrm{site}\}}}(g))$

$(R_n = (\Delta C, \Delta D, \Delta K))$ typed return residual scalar coherence, polarity

cut winding / modular flux declared lattice currents a served answer

Hidden Quotient translation, used only as a dictionary, never as a proof transfer:

Geometry Measure model

discarded coordinate locally null function

(L^2) witness finite-mass tests

restore a point evaluation enlarge the witness, do not change (μ)

missing suprema in $(\Sigma \mathcal{N}_{\{\mathrm{loc}\}})$ localizable-completion failure

Service omission and witness restoration are different choices of observation. Quotienting away

a stage because $(A_{\{\mathrm{sel}\}})$ survived is omission. Adding (Vis) or (K)

because compliance did not descend is restoration.

10. Operator card

Use this at the point of a claim.

Retain (audit / prediction / correction)

governing request and object anchor

answer-selection rule and the spans it used

completion decision, with UNKNOWN allowed

stage identities, order, and actual visibility

active corrections and eligibility window

tool / result dependencies the task still needs

provenance sufficient to replay the record

any coordinate required by a declared future output (B)

Serve (direct service)

the user object

the validated answer unit

nothing else that Lemma S2 has shown to be visibly nonessential for this specimen and witness

Refuse to infer

displayed $\Rightarrow$ internally unused

deleted from the view $\Rightarrow$ deleted from the process

same current answer $\Rightarrow$ same next answer

same route shape $\Rightarrow$ same service class

gate closure $\Rightarrow$ residual closure

cardinality $\Rightarrow$ geometry

model agreement $\Rightarrow$ empirical event

11. What the carried evidence may and may not do

The 9 September ablation packet reports source-linked exact preservation for 287 selected answers, 74 chains with separate stages, 156 separate records, and 137 chain-by-stage-class ledger rows. Those figures license ($A_{\{\mathrm{sel}\}}$)-level class preservation under the supplied deletions. They do not license:

($A_{\{\mathrm{task}\}}$) completeness (located selection failures remain),

live service-route mutation,

a ranking of systems from unequal route samples,

a claim that B50 or 455 is an identified internal pipeline,

pooling with DROP_IT v4.2 or any other denominator.

Correction-binding (129 nonrecurrences / 198 comparable visible opportunities) keeps its own denominator, exclusions, and clustered-opportunity scope. It is a compliance witness, not a service-quotient identity.

No result in this card changes Q1/Q2/Q3 status, the toroidal accepted-event discrepancy, or the Hidden Quotient's standing as packaged classical analysis.

12. Checksum for this successor

Were audit witness and service witness kept distinct?

Was eligibility part of the domain of ($\pi_{\{\mathrm{DS}\}}$)?

Was ($A_{\{\mathrm{sel}\}}$) forbidden to stand in for ($A_{\{\mathrm{task}\}}$) without a completion rule?

Did ablation claims stay inside representation, not execution?

Was (d_{DS}) kept off (Q_{DS}) ?

Was correction scored on (π_{audit}) , not recovered from $((O,A))$?

Was predictive closure of (π_{DS}) treated as false in general, by Theorem P?

Were Hidden Quotient, 39-screen, $(\mathcal{L})$, and (Q_{DS}) left unfused?

Did every empirical figure keep its original denominator?

Does the final sentence stay inside the claim boundary?

Final banner

...

RETAIN WHAT THE WITNESS NEEDS.

SERVE THE OBJECT AND THE ANSWER.

DO NOT MAKE A QUOTIENT DO A WITNESS'S JOB.

...

v2.1 is an append-only successor. It does not rewrite the 39-screen, the Hidden Quotient faces, toroidal winding, or OMNIBUS §10. It gives the service layer exact lemmas and stops them from eating the rest of the geometry.

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